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Beyond the Toolbox: Shaping Technical Leaders Through Cross-Disciplinary Growth and Strategic Exposure

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Abstract

As the energy sector faces unprecedented complexity - driven by digitalization, sustainability transitions, and multidisciplinary integration - the need for technical leadership has become increasingly critical. Technical leadership is not merely an extension of technical expertise nor a traditional management role; rather, it reflects the ability to influence, integrate, and guide others across complex, high-stakes environments. This paper explores how technical leaders evolve, what skills they require, and how organizations can systematically develop them.

The proposed framework outlines a four-level model of technical leadership - Operational, Subject Matter Expert, Integrative, and Strategic - each with distinct behaviors, influence, and development needs. Rather than relying on tenure or narrow expertise, the framework emphasizes early exposure to complexity, cross-functional fluency, and progressive development. A whole-system approach is proposed, supported by five organizational enablers: strategic framing, structured exposure, visible recognition, alternative advancement paths, and governance mechanisms that reinforce leadership behaviors.

The analysis is grounded in literature, field insight, and organizational practice. While technical leadership often begins with individual initiative, scalable impact requires institutional commitment. Organizations that proactively identify, develop, and empower high-potential technical talent will be better positioned to deliver safe, sustainable, and innovative performance. The practical, structured pathway proposed for building such capability.

Introduction

Technical excellence has long been the cornerstone of success in the energy sector, particularly in disciplines such as drilling, subsurface engineering, and field operations. Engineers and specialists are routinely tasked with solving high-risk, high-cost challenges under tight timelines and volatile conditions. However, as the industry continues to evolve - through unconventional resource development, integrated project delivery, digital transformation, and heightened sustainability expectations - success increasingly depends on more than technical competence alone. It requires leadership.

Today's most pressing challenges - such as reducing well delivery time, minimizing non-productive events, optimizing cross-functional workflows, and enabling continuous learning across the asset lifecycle

- cannot be solved by individual experts working in isolation. They require professionals who can integrate knowledge across disciplines, align engineering decisions with business strategy, and influence teams beyond their formal authority. In short, the industry needs technical leaders.

Despite this clear demand, most technical professionals encounter a developmental gap. While their responsibility and visibility may grow over time, their leadership journey often lacks structure, guidance, or intentional design. Many find themselves at a career crossroads: either continue deepening their technical specialization with limited organizational influence or shift toward people management - often without adequate preparation or interest. The result is a persistent gap between technical capability and true leadership readiness.

Compounding this issue is the reality that most corporate leadership programs are not built with technical talent in mind. Traditional models emphasize abstract soft skills, hierarchical authority, or generic business acumen. They rarely account for the unique context of technical professionals operating in field-centric, data-driven, and often safety-critical environments. For example, a drilling superintendent or a geomechanics specialist may lead multimillion-dollar decisions daily, yet remain excluded from formal leadership tracks simply because they do not manage direct reports.

This paper addresses that gap by exploring how technical leaders emerge, what capabilities they must develop, and how organizations can support their growth through structured, practical means. Drawing on field observations, career trajectory mapping, and program design insights, the paper outlines a framework for shaping leadership within technical domains - without forcing engineers to abandon their technical identity.

Rather than promoting a single path, the paper highlights multiple leadership trajectories grounded in field realities - rotational experiences, stretch assignments, cross-functional teaming, and exposure to decision-making forums. It argues for an expanded definition of leadership that values integration, influence, and system thinking, not just management titles.

Ultimately, this paper seeks to help both individuals and organizations answer a critical question: How do we turn today's engineers and specialists into tomorrow's technical leaders - capable of guiding teams, shaping decisions, and driving performance at the intersection of technology, business, and people?

Literature overview

The development of technical leaders in the energy sector has become increasingly important as the industry undergoes rapid digitalization, sustainability transformations, and faces complex interdisciplinary challenges. Across both academic and industry literature, technical leadership is consistently defined as the integration of deep domain expertise with leadership capabilities that extend beyond traditional management roles. These include communication, systems thinking, adaptability, digital fluency, and strategic vision.

Technical leadership extends beyond technical mastery to include the capacity for solving complex problems through holistic thinking and multidisciplinary collaboration. [Schreiner, Gibson, and Morales \(2023\)](#) emphasize systems thinking as foundational, enabling leaders in energy and industrial contexts to manage interconnected systems and foster innovation. Digital leadership research (e.g., [Anamaria, 2023](#); [López-Figueroa et al., 2023](#)) adds competencies such as agility, strategic foresight, and collaborative ability as critical in navigating technological disruption.

Northouse's foundational work (2019 edition) distinguishes technical leaders as those who combine specialized expertise with influence and foresight - qualities that are vital in high-stakes, technically intensive environments like oil & gas and renewables.

The transition from subject-matter expert to technical leader is one of the most critical - and often unsupported - stages of professional development. [Mahlangu and Govender \(2015\)](#) identify significant skill gaps in communication, team coordination, and organizational awareness when technical staff are promoted into leadership roles. [Sharma et al. \(2021\)](#) presented a technical leadership development program

for engineering students that emphasizes systems thinking, collaboration, and communication - highlighting the importance of structured support outside traditional management curricula.

Digital transformation has compounded the complexity of this transition. Leaders are now expected to integrate classical interpersonal leadership skills with digital competencies. Using the Delphi method, [Philip et al. \(2023\)](#) identify behavioral and strategic agility as essential for new technical leaders operating in volatile, tech-driven environments.

One structural solution receiving increasing attention is the dual career ladder model. [Weer and Greenhaus \(2015\)](#) provide a comprehensive discussion of how dual ladders allow for parallel advancement - where technical experts are rewarded and promoted for their contributions without being forced into people-management roles. [Strenalyuk et al. \(2023\)](#) add to this discussion by presenting frameworks for organizational competency mapping that support long-term innovation and talent retention.

[Bartol and Larson \(2020\)](#) address a complementary aspect: the growing importance of technical leaders managing distributed and technology-mediated teams. This further extends the skill set beyond traditional operational leadership to include digital collaboration, influence without proximity, and virtual decision-making.

Sector-specific literature in engineering and energy highlights the need for technical leadership that integrates safety, sustainability, operational excellence, and commercial awareness. [Alharbi \(2022\)](#) provides a comprehensive overview of engineering leadership in the smart manufacturing context, pointing out the lack of global consensus on competencies and the regional skew toward North American and European case studies. [Schreiner et al. \(2023\)](#) reaffirms the importance of systems thinking in risk-sensitive environments and show how project-based learning programs increasingly simulate the complexity of field leadership.

At the macro level, publications from the Harvard Business Review, MDPI, and SPE reflect broader sectoral transformations - decarbonization, digitalization, and cross-disciplinary collaboration - all of which require new forms of technical leadership. These publications emphasize the importance of leadership that spans engineering, environmental stewardship, regulatory engagement, and business alignment.

While literature has expanded, several notable gaps remain. There is still no globally unified set of technical leadership competencies, and most frameworks remain fragmented and locally defined ([Alharbi, 2022](#)). Robust empirical evaluation of leadership development programs is limited, particularly in longitudinal or sector-specific studies ([Sharma et al., 2021](#)). The energy and oil & gas sectors are underrepresented in academic research compared to general engineering domains, and digital competencies - despite their rising importance - are not yet fully integrated into mainstream technical development programs ([Philip et al., 2023](#); [Anamaria, 2023](#)). Furthermore, the mechanisms supporting mid-career transitions - such as structured mentoring, competency mapping, or experiential learning pathways - require much more attention in both research and practice.

In summary, technical leadership in the energy sector demands a distinctive blend of rigorous technical competence and dynamic leadership skills, including systems thinking, communication, adaptability, and digital capability. Career transitions from expert to leader remain complex and require targeted programs, institutional support, and cultural reinforcement. Dual career ladder models offer a promising structural foundation but are not widely implemented or evaluated. Sector-specific leadership, especially in oil & gas, requires further definition, empirical validation, and alignment with emerging priorities such as digitalization and sustainability. Addressing these gaps will be critical to equipping the next generation of technical leaders to operate confidently at the intersection of technology, people, and performance.

Defining technical leadership

As the energy sector becomes increasingly complex and multidisciplinary, the term "technical leadership" is often used but rarely defined with precision. Too often, it is conflated with people management or mistaken for mere technical seniority. Yet, true technical leadership is a distinct and essential capability -

one that bridges deep subject matter expertise with the ability to influence, align, and guide others in high-consequence environments.

Unlike project managers or functional supervisors, technical leaders often operate in roles without direct reports or positional power. Their influence comes from trust, track record, and judgment. For example, a senior drilling engineer might not manage people but may shape the well design philosophy, challenge unsafe proposals, or mentor others in how to navigate difficult formations. A production engineer may not lead a department but could initiate a cross-functional effort to reduce flaring or debottleneck a system. In these roles, technical leadership shows up in decisions made, risks avoided, and standards raised - not in organizational charts.

The behaviors of effective technical leaders tend to cluster around several observable traits. They demonstrate systems thinking - seeing the whole picture, not just their piece of it. They balance technical integrity with execution, adapt without compromise, and integrate across domains. They lead through questions and coaching, elevating others by framing problems clearly. And critically, they take ownership for continuous learning - embedding lessons, challenging outdated practices, and building collective capability.

What distinguishes technical leadership from isolated technical initiative is the potential for scalable impact. In its early form, leadership often emerges in the field - when engineers show good judgment under pressure, solve problems collaboratively, or innovate in execution. But technical leadership does not end at the rig or office desk. It becomes leadership when those contributions evolve into broader influence: shared learning, standardized practice, or organizational change.

This progression reflects a shift from operational initiative to growing leadership influence. An engineer who improves rig performance is a valuable contributor; one who formalizes that learning, scales it across assets, and challenges upstream assumptions is beginning to operate as a technical leader. Their influence crosses boundaries - between disciplines, between functions, and over time, between levels of the organization.

Cross-disciplinary fluency is often the strongest predictor of this evolution. Leaders who understand geology, drilling, fluids, completions, safety, and business constraints engage more effectively across teams. They anticipate unintended consequences, reconcile conflicting inputs, and help reach alignment - not through authority, but through credibility. In contrast, niche expertise - no matter how deep - rarely scales beyond its domain without this broader fluency.

Recognizing technical leadership means looking beyond job titles or technical deliverables. It means identifying those who:

- Operate with a system-level mindset
- Influence decisions beyond their function
- Build alignment across technical and non-technical teams
- Take ownership for continuous improvement
- Elevate others through mentorship and clarity

At its core, technical leadership is the ability to apply specialized knowledge to real-world challenges while shaping outcomes that extend beyond one's individual contribution. It is the bridge between technical depth and organizational direction.

To structure this evolution, a four-level framework is proposed that defines technical leadership by scope of influence and organizational value:

A Four-Level Model of Technical Leadership

Operational Leadership. This is leadership at the point of execution - on the rig, in the control room, or in design reviews. It is grounded in technical credibility and sound decisions under pressure. Engineers at this level lead informally by solving problems, supporting peers, and preventing inefficiencies. They are often the first to be trusted with coaching others, setting the tone for quality and discipline.

Subject Matter Expert (SME). SMEs are recognized for their depth in a specific technical field - be it fluids, casing design, or managed pressure drilling (MPD). Their leadership is based on expertise, reliability, and the ability to solve intricate problems. They shape standards, conduct root cause analyses, and mentor newer engineers. While highly respected, their influence may remain siloed unless intentionally developed further.

Integrative Leadership. Integrators work across domains and silos. They resolve tensions between technical integrity and execution, between commercial and operational needs, or between safety and innovation. These leaders align geology, drilling, Health, Safety, and Environment (HSE), and business strategy into coherent decisions. They often lead cross-disciplinary teams, field development plans, or digitalization initiatives where no single function owns the outcome.

Strategic Technical Leadership. At this level, technical leaders shape enterprise direction. They advise executives on technical risk, influence capital allocation, and represent the company in regulatory or industry forums. Their credibility spans internal and external boundaries. These leaders do not simply challenge designs - they shape the questions, redefine trade-offs, and help the company choose what to solve next.

This framework emphasizes that technical leadership is not a reward for tenure - it is a pathway of growing influence. Companies that understand this distinction are better able to identify and develop technical talent, not just for their expertise, but for their potential to shape the future of the business.

Development Pathways & Skills

Developing technical leaders in the energy sector requires a fundamental shift from traditional, seniority-based models to a structured, inclusive, and strategic development architecture. This chapter proposes a novel, integrated framework designed to build leadership capability in technically skilled professionals - not only by promoting individuals into roles of authority, but by deliberately cultivating cross-disciplinary judgment, influence, and systems thinking as part of their day-to-day trajectory.

The starting point for technical leadership development must be a clear separation between leadership development and positional promotion. In many engineering-centric organizations, leadership capability is often recognized only when it is accompanied by formal authority - such as line management or project ownership. This conflation excludes a large portion of high-potential individuals who may influence outcomes, challenge assumptions, and integrate complexity long before they are promoted. Instead, development should be framed as an organizational learning system, wherein technical professionals are systematically exposed to increasing levels of technical, organizational, and interdisciplinary complexity.

A developmental framework must recognize that technical leadership is not a fixed destination but an evolving capability. Exposure to cross-functional challenges should not be reserved for those already deemed "ready" but used as the very mechanism to build readiness. This includes embedding early-career engineers and mid-career experts in projects or roles that require decision-making under uncertainty, integration across technical domains, and participation in trade-off discussions where technical integrity, operational constraints, and business outcomes intersect. Such exposure builds what can be described as judgment maturity - a critical, often underdeveloped, trait in technical leaders.

Furthermore, growth in technical leadership must be supported through a transparent and structured developmental scaffolding. Rather than relying on isolated training programs or ad hoc mentorship, organizations should establish a consistent set of mechanisms that build integrative capability over time. These may include:

- Structured complexity immersion - planned rotational assignments or co-ownership of interdisciplinary initiatives that deliberately require the candidate to synthesize inputs from multiple domains.
- Contextual decision forums - cross-functional design or risk-assessment reviews where high-potential individuals are invited not to observe, but to present, challenge, and defend options.
- Shadowing and backward mentoring - pairing emerging leaders with senior integrators not only for exposure, but to reverse-explain decisions in structured debriefs, developing both contextual literacy and articulation skills.

An essential aspect of this framework is visualizing development in terms of evolving behaviors rather than fixed roles. The four-stage behavioral trajectory to guide both assessment and development is proposed:



Figure 1—The four-stage behavioral trajectory

Each stage is characterized not only by broader scope, but by increased cognitive integration, abstraction capability, and influence across technical and non-technical domains. Advancement through these stages requires not time-in-role, but deliberate exposure, feedback, and institutional recognition of behaviors aligned with these archetypes.

Crucially, this framework must be inclusive, not selective. Rather than designing development as a pathway for a pre-identified minority of "high-potentials," organizations should implement systems that allow all technical staff to access leadership development signals early and visibly. This creates a merit-based environment in which high-potential individuals naturally emerge, while others may develop at their own pace and still contribute meaningfully. Selection should not be a prerequisite for development; rather, development should inform eventual selection. The goal is not to democratize leadership outcomes, but to democratize access to leadership-building experiences.

Equally important is the role of the organization itself. It is insufficient to place the burden of growth on the individual alone. Organizations must actively curate the development process, not only through HR-led programs but through operational design. For example, assigning early-career engineers to lead technical investigations, involving subject-matter experts in project framing discussions, or appointing experienced

engineers to lead cross-disciplinary reviews without requiring line authority - all represent deliberate acts of developmental design.

Feedback and recognition systems must also evolve. Leadership potential should be assessed based on observed behaviors - such as cross-functional influence, quality of judgment under ambiguity, systems thinking, and mentorship of others - not only on delivery metrics or compliance with KPIs. Performance reviews should include inputs from multiple disciplines, and success stories of non-authority leadership should be made visible throughout the organization to reshape cultural norms.

In sum, the development of technical leaders should be treated as a strategic system, not a discretionary practice. It must combine structured exposure to complexity, clear behavioral archetypes, inclusive access to growth experiences, and deliberate institutional scaffolding. By doing so, organizations in the energy sector can build a robust pipeline of integrative, strategic, and high-impact technical leaders capable of navigating not only technical challenges, but the multifaceted transitions ahead - from digitalization to decarbonization.

Organizational Enablers

The development of technical leaders is not a matter of individual capability alone - it is deeply influenced by the structures, norms, and systems within which they operate. Organizations that consistently cultivate strong technical leaders tend to take a deliberate and strategic approach, creating conditions where leadership potential can emerge, be recognized, and be developed. This chapter outlines five key enablers that, together, represent a whole-system approach to building technical leadership capacity.

Strategic Alignment of Career Tracks

Many organizations claim to support dual career ladders - one for management, one for technical expertise - but in practice, these tracks are often misaligned. Promotions, influence, and access to decision-making are typically funneled through the managerial path, making technical leadership feel like a secondary option. To correct this, companies must create clear, credible, and prestigious technical pathways that lead to enterprise-level impact and recognition. This includes defining what success looks like at each technical level, ensuring parity in rewards and visibility, and assigning enterprise-relevant responsibilities - not just technical depth - to senior technical roles.

Deliberate Exposure to Complexity

Leadership does not emerge in isolation - it develops through interaction with ambiguity, trade-offs, and the need to influence others. For technical professionals, this means being placed in situations that stretch beyond their core expertise: cross-functional projects, high-consequence problem-solving, integration roles, and moments of operational ambiguity. Exposure to such complexity must be intentional, supported by reflection and feedback, rather than left to chance. Organizations should map out developmental rotations, task force roles, and design reviews where technical influence can be observed, supported, and grown.

Recognition Systems That Reflect Integrative Impact

Most performance systems still reward output and individual contribution. However, technical leadership often shows up in behaviors that are hard to measure with traditional KPIs: coaching others, simplifying complex problems, building cross-functional alignment, and raising the standard of thinking. To cultivate this, organizations must revise recognition systems to capture such contributions. This can include peer nominations, qualitative performance signals, and special recognitions for system-level thinking and knowledge transfer. What gets noticed and rewarded sends strong cultural signals about what is valued.

Sponsorship, Not Just Mentorship

While mentorship provides support, sponsorship provides advocacy. Many high-potential technical individuals remain invisible to the broader organization unless someone with influence actively opens

doors for them. Unlike mentorship, which focuses on guidance, sponsorship involves active endorsement: recommending for key roles, assigning complex challenges, and ensuring the individual is seen in leadership contexts. Building a sponsorship model - particularly for underrepresented talent - is a critical enabler of equitable technical leadership development. It also creates a feedback loop where today's leaders are responsible for developing tomorrow's.

Governance and Ownership of the Technical Pipeline

Without formal ownership, technical leadership development becomes fragmented or purely reactive. Organizations need clear governance over their technical talent pipeline - defining who is responsible for identifying potential, tracking growth, assigning opportunities, and maintaining standards of readiness. This governance does not sit in HR alone; it must involve senior technical leaders, business line leaders, and talent partners working together to curate and evolve the technical leadership ecosystem. Dashboards, succession data, and joint reviews of progress can help institutionalize this oversight.



Figure 2—Organizational enablers

These five enablers - strategic alignment of career tracks, deliberate exposure to complexity, recognition of integrative impact, sponsorship of high-potential talent, and structured governance of the technical pipeline - form the infrastructure through which technical leadership can thrive. They were selected because they represent a whole-system approach: from strategy → exposure → recognition → advancement → system oversight. Without such coordinated mechanisms, even the most capable individuals may never reach their full leadership potential.

Conclusion

Technical leadership in the energy sector is not simply an extension of individual technical expertise, nor is it interchangeable with people management. Rather, it reflects a distinct capability: the ability to apply deep domain knowledge while influencing others, guiding decisions, and integrating across disciplines in high-consequence, uncertain environments. This form of leadership is exercised through credibility, judgment, and collaboration -not hierarchy.

A four-level developmental model was proposed to articulate the progression of technical leadership: Operational, Subject Matter Expert, Integrative, and Strategic. Each level represents a shift in scope, systems thinking, and influence, providing a structured way to identify and support leaders as they grow. Notably, these levels are not tied to years of seniority or formal authority, but to demonstrated behaviors and integrative capability.

Leadership development in technical domains cannot rely solely on time or informal exposure. It requires intentional cultivation - starting early and continuing across the career trajectory. Exposure to complexity, structured opportunities to operate beyond one's discipline, and recognition of systems-level contributions are critical. High-potential individuals benefit most from these environments, but the opportunity for leadership development should be broadly accessible, not artificially restricted.

At the organizational level, sustained progress depends on enablers that go beyond ad hoc training programs. Five levers were identified as essential for building a robust leadership pipeline:

1. Strategic alignment of technical and managerial career tracks
2. Deliberate exposure to cross-disciplinary complexity
3. Recognition systems that value technical influence
4. Sponsorship and advancement mechanisms for high-potential individuals
5. Governance over the leadership development pipeline

These levers were selected because they represent a whole-system approach: from long-term strategy to day-to-day decision-making, from early-career exposure to enterprise-level succession planning. When activated together, they form the foundation of a leadership culture that supports growth, resilience, and innovation.

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